

Courant Algebroid Relations and T-duality

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Outline

- 1 Sigma-models and Courant algebroids
- 2 T-duality and Relations
- 3 Example
- 4 Outlook

Courant algebroids

The **generalised tangent bundle**, $TM \oplus T^*M$, can be equipped with a pairing and a bracket, $H \in \Omega_{\text{cl}}^3(M)$:

$$\begin{aligned}\langle X + \xi, Y + \eta \rangle &= \iota_X \eta + \iota_Y \xi, \\ [X + \xi, Y + \eta]_H &= [X, Y] + \mathcal{L}_X \eta - \iota_Y d\xi + \iota_Y \iota_X H.\end{aligned}$$

In general, a **(exact) Courant algebroid** (CA) is a vector bundle E over M with a pairing and a bracket of sections fitting the sequence

$$0 \rightarrow T^*M \rightarrow E \rightarrow TM \rightarrow 0. \quad (E \cong TM \oplus T^*M)$$

CAs are classified by their *Ševera* class $[H] \in H_{\text{cl}}^3(M)$, with H given by a splitting $s: TM \rightarrow E$

$$H(X, Y, Z) = \langle [s(X), s(Y)]_E, s(Z) \rangle_E.$$

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Wess-Zumino functional

Consider maps f from a worldsheet surface Σ to a target space M .

Lagrangian formulation	Geometry
Wess-Zumino term $S_H[f] = \int_V f^* H$	Courant algebroid E_H over M
$\delta S_H = 0$	$X + \xi$ such that $\iota_X H = d\xi$
Polyakov term $S_g[f] = \int_\Sigma \ df\ _g^2 d\mu_g$	Generalised metric $V_+ \subset E_H$ equiv. $\tau \in \text{End}(E_H)$ equiv. $g \in \text{Riem}(M)$
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Polyakov term and Isomorphisms

We can also add dynamical data to our sigma model: the Polyakov functional, for a Riemannian metric g on M is

$$S_g[f] = \int_{\Sigma} \|df\|_g^2 d\mu_g.$$

The data of g can be captured in a **generalised metric**; a positive definite subbundle $V \subset E$, i.e. $\langle V, V \rangle > 0$. For example,

$$V = \text{gr}(g) = \{X + g(X) : X \in TM\} \subset TM \oplus T^*M$$

The data of the full action $S = S_H + S_g$ is embedded in a CA equipped with a positive definite subbundle V .

Thus, an isomorphism $\Phi: E \rightarrow E'$ covering a diffeomorphism $\phi: M \rightarrow M'$ and preserving $\langle, \rangle_E, [,]_E$ and V will yield

Sigma-modles over M and M' with the same equations of motion.

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Courant algebroid relations

T-duality is an equivalence of Sigma-models on different target manifolds M and M' , though these are not necessarily diffeomorphic. A CA isomorphism $\Phi: E \rightarrow E'$ must cover a diffeomorphism, so we seek to generalise the notion of isomorphism.

Consider the graph $\text{gr}(\Phi) \subset E \times E'$. One sees that

$$[\Phi \cdot, \Phi \cdot]_{E'} = \Phi([\cdot, \cdot]_E) \iff \text{gr}(\Phi) \text{ is involutive in } E \times E'$$

$$\langle \Phi \cdot, \Phi \cdot \rangle_{E'} = \langle \cdot, \cdot \rangle_E \iff \text{gr}(\Phi) \text{ is isotropic in } E \times E'$$

A **CA relation** $R: E \dashrightarrow E'$ is an isotropic, involutive subbundle of $E \times E'$, supported on a submanifold $C \subset M \times M'$.

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Isometry and Composition

To preserve the structure of a generalised metric $V \subset E$, one sees that V is the $+1$ eigenbundle of a $\tau \in \text{Aut}(E)$. If $R: (E, V) \dashrightarrow (E', V')$, then R is a **generalised isometry** if

$$(\tau \times \tau')(R) = R.$$

Contingent on some smoothness conditions, one can compose two relations $R: E \dashrightarrow E'$, $\tilde{R}: E' \rightarrow E''$

$$\tilde{R} \circ R = \{(e, e'') : (e, e') \in R \text{ and } (e', e'') \in \tilde{R}\} \subset E \times \overline{E''},$$

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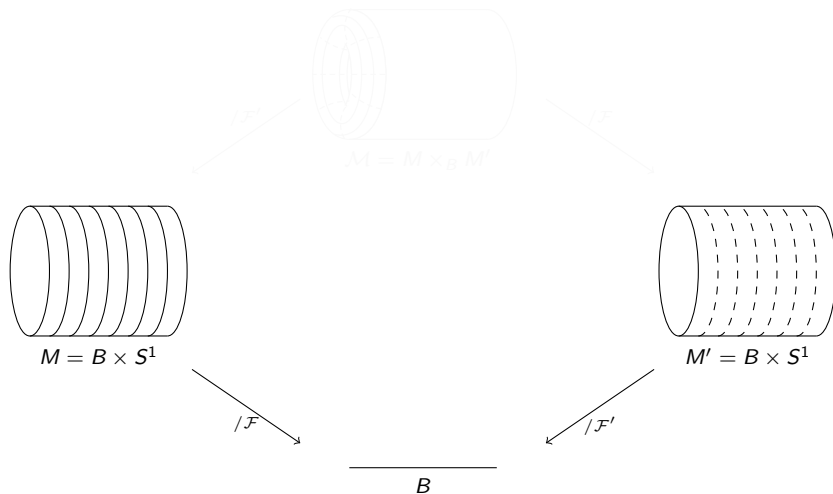
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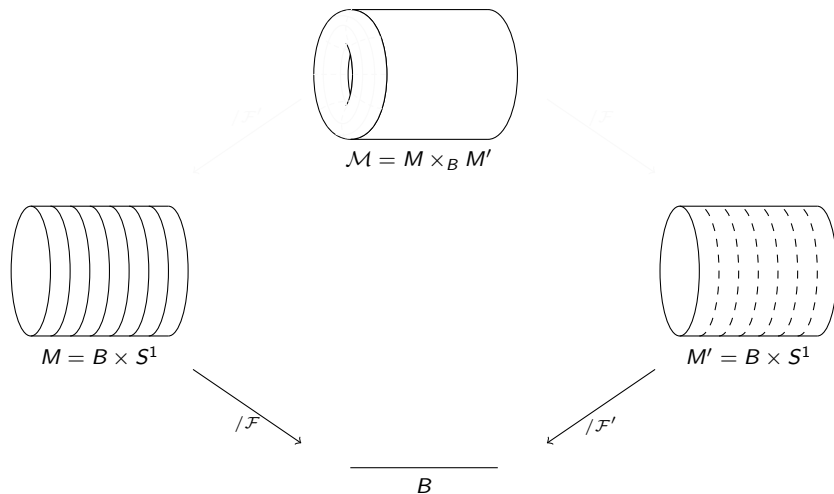
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T-dual spaces M, M' may be fibre bundles over a common base B



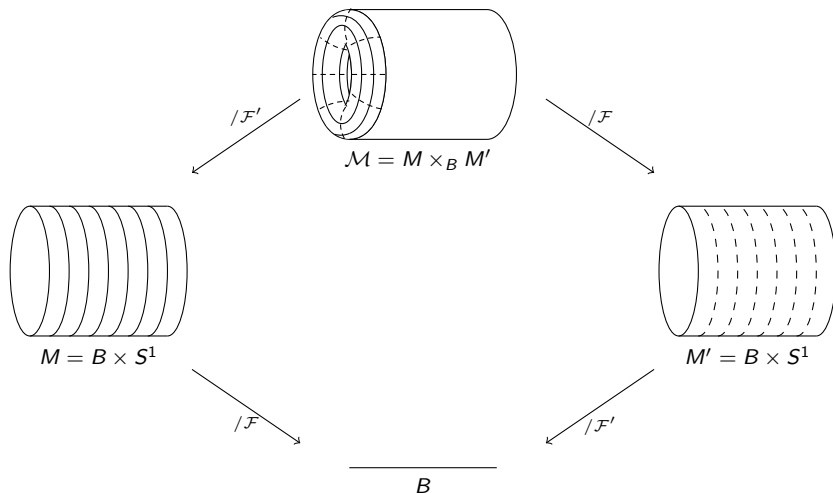
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T-dual spaces M, M' may be fibre bundles over a common base B



The T-duality relation I

Want to form a CA relation $R: E \dashrightarrow E'$ between CAs E, E' over M and M' respectively. Need to know how to form CAs on a quotient manifold:

$\mathcal{M}$ is foliated by $\mathcal{F}'$. If $\mathcal{E}$ is an exact CA over $\mathcal{M}$, then $\mathcal{E}/\mathcal{F}'$ will not be an exact CA over $M = \mathcal{M}/\mathcal{F}'$.

Take $K' = T\mathcal{F}' \subset \mathcal{E}$, then

$$E = \frac{K'^{\perp}}{K'} / \mathcal{F}'$$

is an exact CA over M .

Can form the CA relation $Q(K')$, supported on $\text{gr}(q')$,

$$Q(K') = \{(e, \natural'(e)): e \in K'^{\perp}\} \subset \mathcal{E} \times E$$

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The T-duality relation $R: E \rightarrow E'$ is then the composition

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 \uparrow Q(K')^T \text{ (dashed)} & & \downarrow Q(K) \text{ (dashed)} \\
 E & \xrightarrow{\quad R \quad} & E'
 \end{array}$$

Theorem (DF, Marotta, Szabo [1])

Let V be a generalised metric on E . (Modulo invariance conditions) TFAE

- ① $K^\perp \cap \Phi(K') \subseteq K$.
- ② There exists a unique generalised metric V' on E' such that R is a generalised isometry.

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The Hopf fibration and its T-dual

Consider $\mathcal{M} = S^3 \times S^1$, with a foliation $\mathcal{F}$ generated by the trivial fibres of S^1 , and a foliation $\mathcal{F}'$ with whose leaves are the S^1 fibres of the Hopf fibration $S^3 \rightarrow S^2$. Thus $M = \mathcal{M}/\mathcal{F}' = S^2 \times S^1$ and $M' = S^3$. Let $\partial_\theta, \partial_{\theta'}$ generate $\mathcal{F}, \mathcal{F}'$ respectively.

Let $\mathcal{E} = T\mathcal{M} \oplus T^*\mathcal{M}$. To choose $\Phi \in \text{Iso}(\mathcal{E})$, let θ, θ' be connections on $\mathcal{M} \rightarrow M' = S^3$ and $\mathcal{M} \rightarrow M = S^2 \times S^1$ respectively. Let $B = \theta \wedge \theta' \in \text{Hom}(TP, T^*P)$. Then

$$\Phi(X + \xi) = X + B(X) + \xi.$$

Consider the element $w\partial_\theta + p\theta \in TM \oplus T^*M$, $w, p \in \mathbb{R}$. The relation R is then given by

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Buscher rules and topology

Take a Riemannian metric g on $M = S^2 \times S^1$, that has $g_0\theta\theta$ as the fibre component. Then $V = \text{gr}(g) = \{w\partial_\theta + wg_0\theta\} \oplus \text{gr}(g_{S^2})$.

Thus, the subbundle V' is

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One can also calculate the topological terms: If

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Outlook

- ① Looking at Poisson-Lie T-duality, where fibres $\mathcal{F}$ are possibly non-abelian groups.
- ② Add more structure: divergence (or CA connections), allowing to incorporate the dilaton, generalised Ricci tensors and Ricci flow; generalised complex structures, allowing incorporation of branes.

End

Thanks you for listening! Questions?

References: [1] <https://arxiv.org/abs/2308.15147>